VIT-AP UNIVERSITY

DIGITAL LOGIC DESIGN

(Course Code: ECE 1003)

Module-2:Lecture-8

Combinational Circuit Design

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Module-2

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Lecture-8

- ❖ Magnitude Comparator

BOOKS

Textbooks

1. M.Morris Mano, Michael D Ciletti, **Digital Design**, 5th edition, Pearson Publishers, 2013.
2. R.P. Jain, "**Modern Digital Electronics**", 4th edition, TMH.

References

1. M.Morris Mano, Charles R. Kime, Tom Martin, **Logic and Computer Design Fundamentals**, 4th edition, Pearson Publishers.
2. C. H. Roth and L. L. Kinney, **Fundamentals of Logic Design**, 5th edition, Cengage Publishers.

Magnitude Comparator

MAGNITUDE COMPARATOR

- ✓ A **magnitude digital Comparator** is a **combinational circuit** that compares two digital or binary numbers in order to find out whether one binary number is **equal, less than or greater than** the other binary number.
- ✓ We logically design a circuit for which we will have two inputs one for A and other for B and have three output terminals, one for **A > B** condition, one for **A = B** condition and one for **A < B** condition.

MAGNITUDE COMPARATOR

1-bit Comparator

- ✓ A comparator used to compare two bits is called a **single bit comparator**.
- ✓ It consists of two inputs each for **two single bit numbers** and three outputs to generate **less than, equal to and greater than** between two binary numbers.
- ✓ From the truth table, the **Boolean expressions** can be obtained as

(A>B) = AB’

(A=B) = A’B’ + AB

(A<B) = A’B

A	B	A>B	A=B	A<B
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

MAGNITUDE COMPARATOR

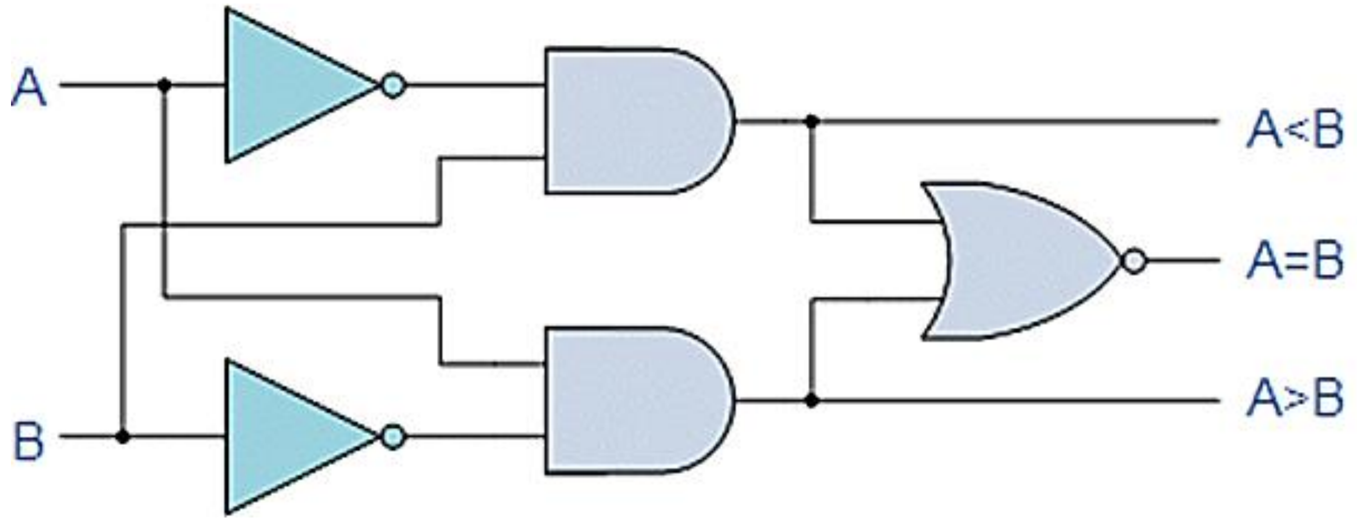
1-bit Comparator

- ✓ The circuit for 1 bit comparator circuit can be designed as

(A>B) = AB’

(A=B) = A’B’ + AB

(A<B) = A’B



MAGNITUDE COMPARATOR

2-bit Comparator

- ✓ A comparator used to compare **two binary numbers each of two bits** is called a **2-bit Magnitude comparator**.
- ✓ It consists of **four inputs and three outputs** to generate **less than, equal to and greater than** between two binary numbers.
- ✓ The truth table for a 2-bit comparator is given as:

Inputs				Outputs		
A_1	A_0	B_1	B_0	A>B	A=B	A<B
0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	1	0	0	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	1	0

2-bit Comparator

✓ The Boolean functions for **A>B** can be obtained as

A_1A_0 \ B_1B_0	00	01	11	10
00				
01	1			1
11	1	1		1
10	1	1		

$A_0B_1'B_0'$

$A_1A_0B_0'$

A_1B_1'

$$(A > B) = A_1B_1' + A0B_1'B_0' + A1A0B0'$$

MAGNITUDE COMPARATOR

2-bit Comparator

✓ The Boolean functions for **A=B** can be obtained as

A_1A_0 \ B_1B_0	00	01	11	10
00	**1**			
01		**1**		
11			**1**	
10				**1**

$$(A = B) = A_1'A_0'B_1'B_0' + A_1'A_0B_1'B_0 + A_1A_0B_1B_0 + A_1A_0'B_1B_0'$$

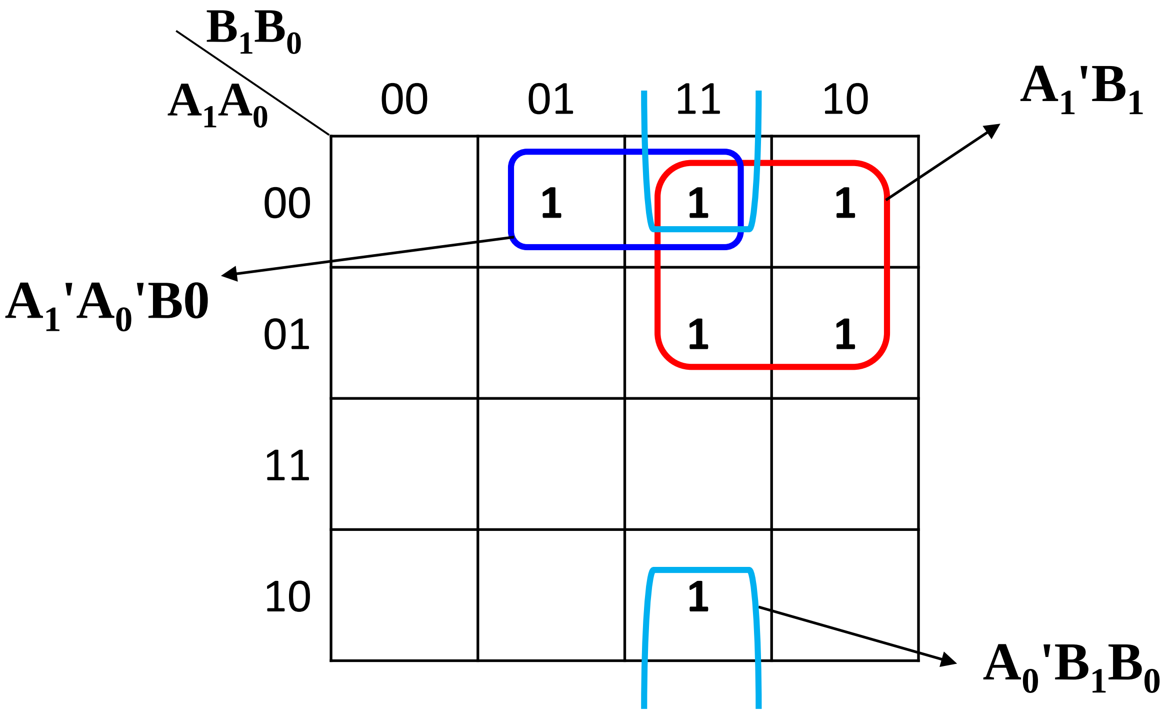
$$= A_1'B_1'(A_0'B_0' + A_0B_0) + A_1B_1(A_0B_0 + A_0'B_0')$$

$$= (A_0B_0 + A_0'B_0')(A_1B_1 + A_1'B_1')$$

$$= (A_0 \odot B_0)(A_1 \odot B_1)$$

2-bit Comparator

✓ The Boolean functions for **A<B** can be obtained as

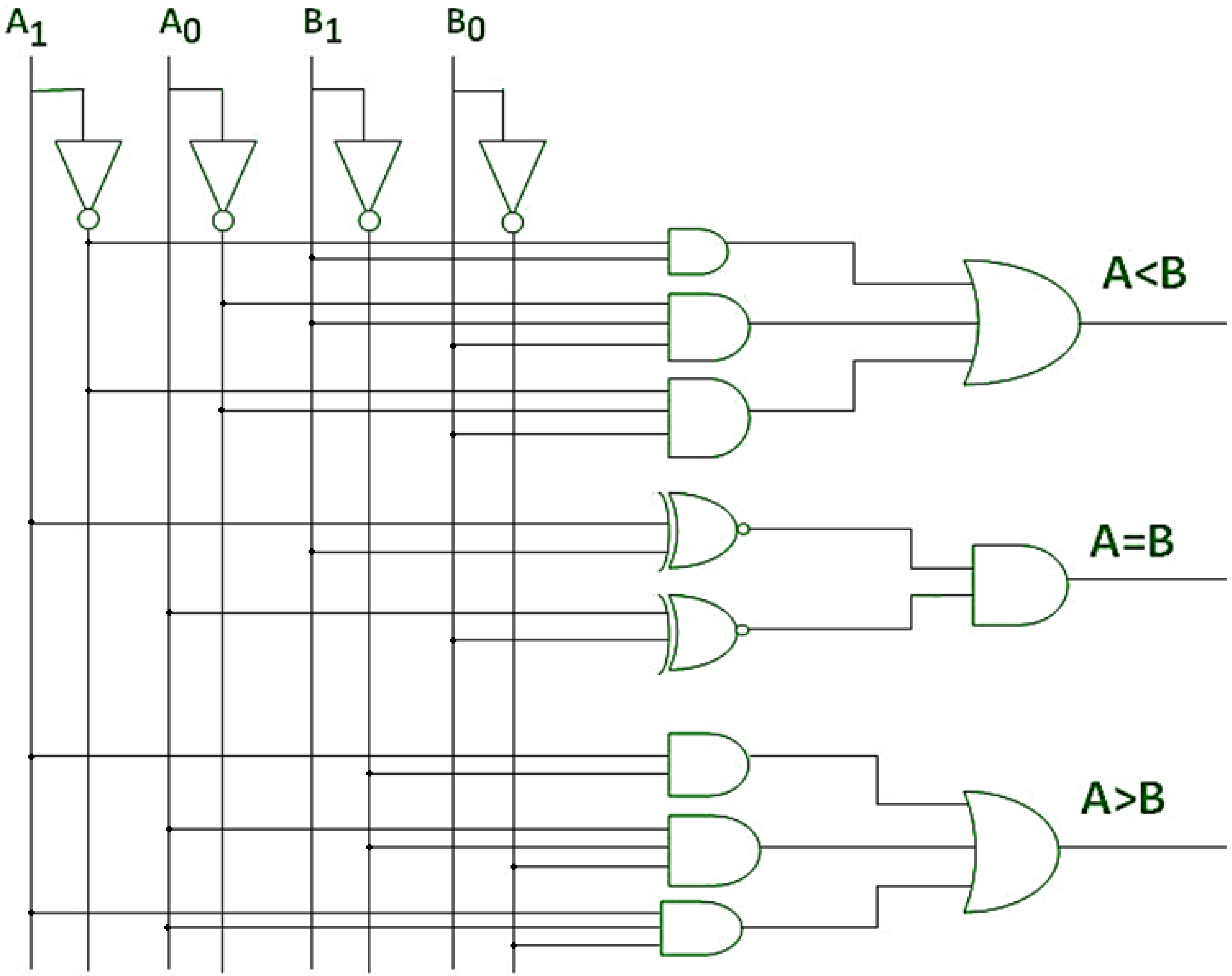


$$(A > B) = A_1'B1 + A0'B_1B_0 + A1'A_0'B0$$

MAGNITUDE COMPARATOR

2-bit Comparator

✓ The Circuit for A<B, A=B and A>B can be designed as



MAGNITUDE COMPARATOR

4-bit Comparator

- ✓ A comparator used to compare **two binary numbers each of 4 bits** is called a **4-bit Magnitude comparator**.
- ✓ It consists of **eight inputs and three outputs** to generate **less than, equal to and greater than** between two binary numbers.
- ✓ In this case, the eight inputs will form **$2^8 = 256$ possible input combinations** which is very difficult to analyze with truth table.
- ✓ Hence, a **direct Boolean function method** can be adopted in order to design the combinational circuit.
- ✓ Consider two numbers, A and B, with four digits each. Write the coefficients of the numbers in descending order of significance:

$$A = A_3\,A_2\,A_1\,A_0$$
$$B = B_3\,B_2\,B_1\,B_0$$

4-bit Comparator

1. $A = B$ Condition:

- ✓ The two numbers are equal if all pairs of significant digits are equal: $A_3 = B_3$, $A_2 = B_2$, $A_1 = B_1$, and $A_0 = B_0$.
- ✓ When the numbers are binary, the digits are either 1 or 0, and the equality of each pair of bits can be expressed logically with an exclusive-NOR function as

$$\boxed{x_i = A_iB_i + A_i'B_i'} \quad \text{for } i = 0, 1, 2, 3$$

where $x_i = 1$ only if the pair of bits in position i are equal (i.e., if both are 1 or both are 0).

- ✓ For equality to exist, all x_i variables must be equal to 1, a condition that dictates an AND operation of all variables:

$$\boxed{(A = B) = x_3x_2x_1x_0}$$

The binary variable ($A = B$) is equal to 1 only if all pairs of digits of the two numbers are equal.

4-bit Comparator

2. $A > B$ Condition:

✓ There are 4 possible cases through which we can analyze ***A>B*** condition

Case-1: If $A_3 = 1$ and $B_3 = 0$, then $A>B$

✓ We can write the Boolean expression as

$$(A>B) = A_3B_3'$$

$$\begin{array}{cccccc} & & A_3 & & & \\ \left.\begin{array}{c} A \\ B \end{array}\right. & \begin{array}{c} \rightarrow \\ \rightarrow \end{array} & \begin{array}{c} 1 \\ 0 \end{array} & \begin{array}{c} 1 \\ 1 \end{array} & \begin{array}{c} 0 \\ 0 \end{array} & \left.\begin{array}{c} 0 \\ 1 \end{array}\right\} \quad A > B \\ & & B_3 & & & \end{array}$$

✓ The expression returns 1 if the above case occurred.

MAGNITUDE COMPARATOR

4-bit Comparator

2. $A > B$ Condition:

$$\begin{array}{cccccc} & & A_3 & A_2 & & \\ A & \rightarrow & 1 & 1 & 0 & 0 \\ B & \rightarrow & 1 & 0 & 0 & 1 \\ & & B_3 & B_2 & & \end{array} \Bigg\} \quad A > B$$

Case-2: If $A_3 = B_3$, $A_2 = 1$ and $B_2 = 0$, then $A>B$

- ✓ We can write the Boolean expression for $A_3 = B_3$ as

$$(A=B) = A_3'B_3' + A_3B_3 = x_3$$

- ✓ Similarly, Boolean expression for $A_2 = 1$ and $B_2 = 0$ as

$$(A>B) = A_2B_2'$$

- ✓ The final expression can be obtained by performing the AND operation between both the conditions

$$(A>B) = x_3A_2B_2'$$

4-bit Comparator

2. $A > B$ Condition:

Case-3: If $A_3 = B_3$, $A_2 = B_2$, $A_1 = 1$ and $B_1 = 0$, then $A>B$

	A_3	A_2	A_1		
A →	1	0	1	0	A > B
B →	1	0	0	1	
	B_3	B_2	B_1		

- ✓ We can write the Boolean expression for $A_3 = B_3$ and $A_2 = B_2$ as

$$(A=B) = x_3x_2$$

- ✓ Similarly, Boolean expression for $A_1 = 1$ and $B_1 = 0$ as

$$(A>B) = A_1B_1'$$

- ✓ The final expression can be obtained by performing the AND operation between all the conditions

$$(A>B) = x_3x_2A_1B_1'$$

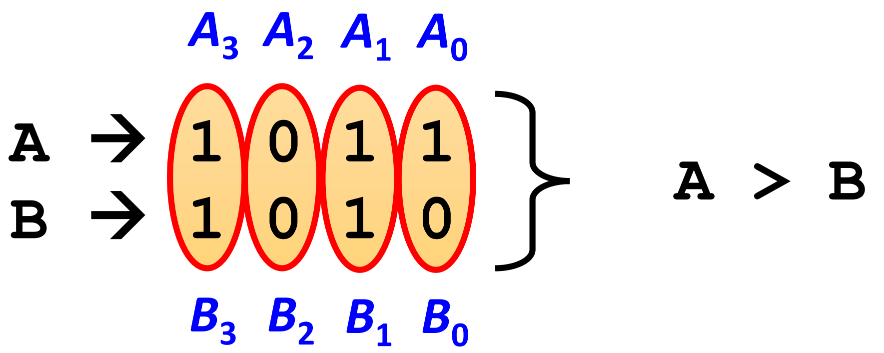
4-bit Comparator

2. $A > B$ Condition:

Case-4: If $A_3 = B_3$, $A_2 = B_2$, $A_1 = B_1$, $A_0 = 1$ and $B_0 = 0$, then $A>B$

- ✓ We can write the Boolean expression for $A_3 = B_3$, $A_2 = B_2$ and $A_1 = B_1$ as

$$(A=B) = x_3x_2x_1$$



- ✓ Similarly, Boolean expression for $A_0 = 1$ and $B_0 = 0$ as

$$(A>B) = A_0B_0'$$

- ✓ The final expression can be obtained by performing the AND operation between all the conditions

$$(A>B) = x_3x_2x_1A_0B_0'$$

MAGNITUDE COMPARATOR

4-bit Comparator

2. *A* > *B* Condition:

- ✓ The final expression can be obtained by performing the AND operation between all the cases

$$(A>B) = A_3B_3' + x_3A_2B_2' + x_3x_2A_1B_1' + x_3x_2x_1A_0B_0'$$

3. *A* < *B* Condition:

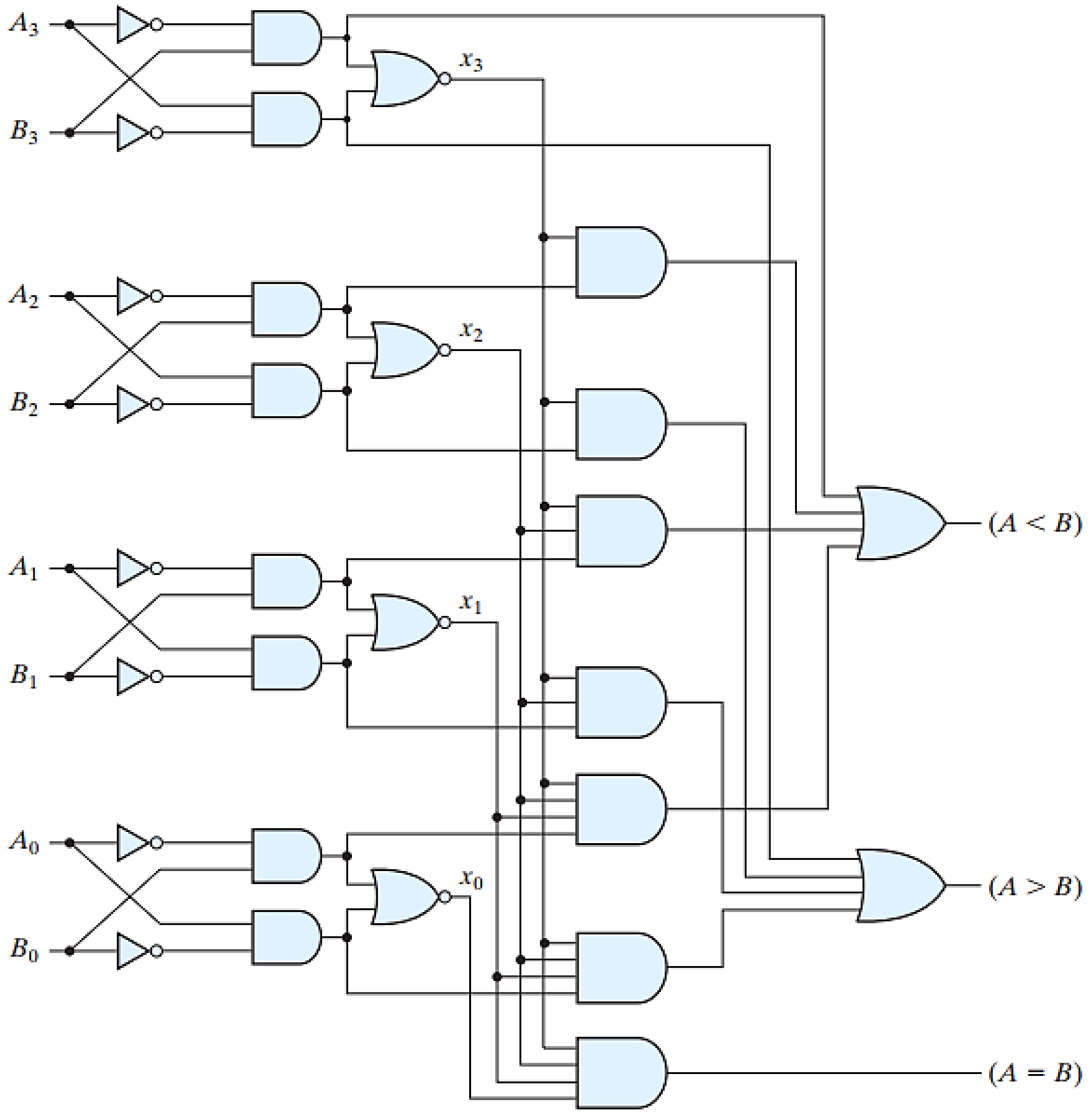
- ✓ Similarly, there are 4 possible cases through which we can analyze ***A<B*** condition.
- ✓ The final expression can be obtained by performing the AND operation between all the cases

$$(A>B) = A_3'B_3 + x_3A_2'B_2 + x_3x_2A_1'B_1 + x_3x_2x_1A_0'B_0$$

MAGNITUDE COMPARATOR

4-bit Comparator

- ✓ The Circuit for A<B, A=B and A>B can be designed as



Thank You